



Faculty of Science and Arts
Department of Chemistry
Biochemistry Lab (CHEM 266)

Experiment 11

Electrophoresis (Sodium dodecyl sulfate Polyacrylamide gel (SDS PAGE))

Purpose: Separate a mixture of Proteins depending on their molecular weight

Prelab questions

- 1- The gels are stained with which stain?
- 2- Are the method used for separation of DNA the same as the one used for protein separation? What are the differences
- 3- Mention three application for this technique?

Precautions

acrylamide in its liquid form is a neurotoxin, and, as such, is not suitable for classroom use. The gels will be provided ready to use.

We will use high voltages to run the gels be aware of electricity hazards, specially that a buffer solution will be used to run the gels

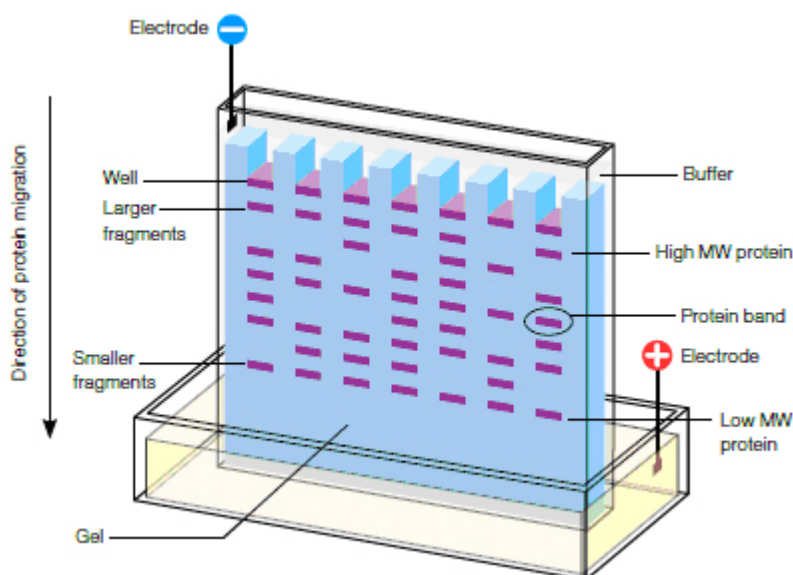
Introduction:

Electrophoresis is an important method for separation and identification of proteins and nucleic acids it's based on migration of charged proteins in an electric field. separation of proteins could be carried out on Cellulose Acetate, Agarose, Sodium dodecyl sulfate Poly acrylamide gel SDS PAGE.

SDS-PAGE, a simple and inexpensive method for resolving proteins in complex mixtures, SDS-PAGE gels provide the starting materials for protein separation , purification or identification , as proteins can be visualized and separated as well as allowing the worker to quickly estimate the number of proteins in a mixture or the degree of purity of a particular protein preparation. It also allows the determination of crucial properties of proteins such as its isoelectric point and approximate molar mass

In SDS-PAGE, an electric field is used to move charged molecules through a matrix of a polymerized acrylamide. Protein across polyacrylamide gels through a networks of. Migration (movment) of proteins across the gels depends on the diameter of the gel pores as well as the size and shape of the molecule. the size of the pore can be

controlled by adjusting the concentration range of the acrylamide. In general, smaller, more highly charged molecules migrate more rapidly through gels than larger or less charged molecules. The mobility of a molecule is also affected by the buffer system and the strength of the electrophoretic field used for the separation.



Electrophoresis Assembly

<http://www.bio-rad.com/en-uk/applications-technologies/protein-electrophoresis-methods?ID=LUSOW4GRI>

The rates at which individual molecules move through the gel depend on the properties of both the separation system and the molecules themselves.

- 1- The size of the molecule. smaller, more highly charged molecules migrate more rapidly through gels than larger or less charged molecules.
- 2- Concentration of the gel material
- 3- The strength of the electric field about 300 volts.

Steps of electrophoresis:

- 1- Before electrophoresis Proteins are denatured , in order to resolve the proteins in a sample In SDSPAGE, proteins are denatured by boiling them with the anionic detergent, sodium dodecyl sulfate (SDS) and 2-mercaptoethanol. The combination of heat and detergent is sufficient to break the many non-covalent bonds that stabilize protein folds, and 2-mercaptoethanol ,SO that when proteins are run on the gel they run; separated according to their molecular weight only,

because after proteins are treated with the SDS they are returned back to their primary structure and they now convey a negative charge approximate to their weight

- 2- Application of the sample.
- 3- Passing the electric current.
- 4- Fixation of protein by an acetic acid and methanol.
- 5- Staining of proteins by Commassie Brilliant Blue
- 6- De staining the gel material by a mixture of Acetic Acid, Water, and Methanol.

** Note that Choosing the buffer depending on the isoelectric point of the protein:

At $\text{pH} > \text{PI}$, the protein carries a net negative charge.

At $\text{pH} < \text{PI}$, the protein carries a net positive charge.

Negatively charged proteins move towards the anode (+) and the positively charged proteins move towards the cathode (-).

References

https://www.nvcc.edu/biotech/_docs/Copy-of-Copy-of-Protein_Electrophoresis_Manual.pdf

<http://capricorn.bc.edu/bi204/wp-content/uploads/2015/08/Chapter-14-2015.pdf>